Autoimmunity Reviews xxx (2017) xxx–xxx

Contents lists available at ScienceDirect

Autoimmunity Reviews

journal homepage: www.elsevier.com/locate/autrev

Using multi-omics methods to understand dermatomyositis/polymyositis☆,☆☆

Siming Gao [a], Hui Luo [a], Huali Zhang [b], Xiaoxia Zuo [a], Li Wang [b], Honglin Zhu [a,*]

[a] Department of Rheumatology, Xiangya Hospital, Central South University, 87 Xiangya Road, Changsha, Hunan 410008, People's Republic of China
[b] Department of Pathophysiology, Xiangya School of Medicine, Central South University, 110 Xiangya Road, Changsha, Hunan 410008, People's Republic of China

ARTICLE INFO

Article history:
Received 2 July 2017
Accepted 8 July 2017
Available online xxxx

Keywords:
Dermatomyositis
Polymyositis
Pathogenesis
Multi-omics

ABSTRACT

Idiopathic inflammatory myopathies (IIM) are a group of rare and heterogeneous autoimmune diseases, and the most common subtypes are dermatomyositis (DM) and polymyositis (PM). Despite extensive efforts, the underlying mechanism of IIM remains unclear. Recent efforts to understand the pathogenesis of IIM have included genomics, epigenetics, transcriptomics, proteomics and autoantibody studies. This review focuses on recent studies in DM/PM research based on multi-omics. This integrated analysis of multi-omics profiling will provide useful insights into DM/PM pathogenesis and recommendations for therapeutic targets and biomarkers development.
© 2017 Published by Elsevier B.V.

Contents

1. Introduction . . . 0
2. Genetics basis of DM/PM . . . 0
3. Epigenetics of DM/PM. . . . 0
4. Transcriptomic profiles in DM/PM . . . 0
5. Proteomics in DM/PM. . . . 0
6. Autoantibodies in DM/PM . . . 0
7. Discussion . . . 0
References . . . 0

1. Introduction

Idiopathic inflammatory myopathies (IIM) are a group of rare and heterogenous autoimmune diseases characterized by progressive symmetric muscle weakness, elevated serum levels of muscle enzymes, electromyographic abnormalities, and inflammatory infiltrates observed upon muscle biopsy [1]. The main clinical subtypes of IIM are classified as dermatomyositis (DM; juvenile, adult), polymyositis (PM), sporadic inclusion body myositis (sIBM) and immune-mediated necrotizing myopathy (IMNM) [2]. The specific clinical feature of DM is the subacute onset of proximal symmetric weakness with a characteristic skin rash in patients of any age, while PM is typified by a subacute proximal myopathy in adults who do not have a rash (diagnosis is made when other causes have been ruled out) [3]. The pathogenesis of DM/PM is not fully understood. Both immune and nonimmune mechanisms are involved in the development of muscle weakness. In DM, interferon (IFN)-producing cells and the increased expression of IFN-induced proteins in both skin and muscle tissues may cause tissue and blood vessel damage. PM is generally regarded as a prototypic T cell-mediated autoimmune myopathy [4–6].

Over the past decade, great efforts have been made to understand the aetiopathogenesis of IIM [7]. High-throughput technologies and integrative systems biology approaches have been widely used in DM/PM, which have greatly improved our understanding of the underlying mechanisms. In this paper, we will systemically review the genomics, transcriptomics, epigenetics, proteomics and autoantibody profiles of DM/PM and discuss how new approaches and technologies are

☆ Sources of funding: This work was supported by grants from the National Natural Science Foundation of China (81373206, 81401357, 81671622), and Independent Innovation Projects of Central South University (2016zzts131).
☆☆ Conflict of interest statement: The authors declare no competing financial interests.
* Corresponding author.
E-mail address: zhuhonglin1988@hotmail.com (H. Zhu).

http://dx.doi.org/10.1016/j.autrev.2017.07.021
1568-9972/© 2017 Published by Elsevier B.V.

broadening our understanding of the pathogenesis of DM/PM and providing important clues for valuable biomarkers.

2. Genetics basis of DM/PM

High-throughput genetic approaches have enabled us to better understand the genetic architecture of DM/PM. The major histocompatibility complex (MHC), also known as the human leukocyte antigen (HLA) region, is regarded as the strongest genetic risk factor in autoimmune disease. The 8.1 ancestral haplotype (8.1 AH) was the first to be identified and has the most variants associated with IIM [8,9]. The strong linkage disequilibrium within this haplotype indicates that multiple genes are strongly associated with IIM. However, which gene or genes contribute to the pathogenesis is not clear.

The strongest association within the MHC region was identified in different populations, and every population was shown to have unique risk HLA haplotypes in IIM. For example, HLA-DRB1*0301, HLA-DQA1*0501 and HLA-A*6802 are involved in White Americans [10], HLA-DRB1*03, HLA-DQA1*05, HLA-DQB1*02 and HLA-DPB1*0101 in UK Caucasians [11,12], and HLA-DRB1*08:03 in Japanese [13]. When stratifying cohorts by clinical subgroups, PM and DM also had common and special risk HLA haplotypes, such as HLA-A*68 in DM and HLA-Cw*14 and HLA-DQA1*0201 in PM in Caucasians [14], HLA-DQA1*0601 in DM in Americans [10], and HLA-DQA1*01:04 and HLA-DRB1*07 in DM in Chinese [15]. HLA-DRB1*03:01 and HLA-B*08:01 were also risk haplotypes associated with DM and PM in Caucasians [8,9,14,16].

The risk HLA haplotypes were also found to have strong associations with the IIM serotype [17]. For example, HLA-DRB1*0301, HLA-B*08, HLA-DPB1*0101, HLA-DQA1*03 and HLA-DQA1*0401 were associated with anti-Jo-1 autoantibodies [10,12,18]. HLA-DRB1*0310-DQA1*0501 was a risk factor for anti-Ro autoantibodies [10]. HLA-DRB1*0302, HLA-DRB1*0701, HLA-DRB1*04, HLA-DQA1*03, HLA-DRB1*07-DQA1*02-DQB1*02, HLA-DQA1*0401 and HLA-DQB1*02 were associated with anti-Mi-2 autoantibodies [10,11,18]. HLA-DRB1*03, HLA-DQA1*0501 and HLA-DQB1*02 were risk factors for anti-PM-Scl autoantibodies [11]. The HLA-DRB1*03 -DQA1*05-DQB1*02 haplotype was a risk factor for anti-aminoacyl-tRNA synthetase (ARS) autoantibodies [11]. HLA-DRB1*1104, HLA-DRB1*0301 and HLA-DQA1*0501 were associated with anti-Ku autoantibodies [17,19]. HLA-DQA1*0102, HLA-DQA1*0104 and HLA-B*5001 were risk factors for anti-SRP autoantibodies [10,17]. HLA-DQA1*06, HLA-DRB1*08, HLA-C*0304 and HLA-DRB1*0301 were associated with anti-La [17], anti-RNP [10], anti-PL-7 and anti-PL-12 autoantibodies, respectively [17]. The HLA-DRB1*03-DQA1*05-DQB1*02 haplotype was strongly associated with interstitial lung disease (ILD) [11].

Many non-HLA susceptibility genes were also identified, such as TNF-α and IKBL (NFKBIL1) in UK Caucasians [20,21], PTPN22 in British Caucasians [22], ACTN3 in Mexicans [23], BLK and STAT4 in Japanese [24,25], IRF5 in Chinese [26], and immunoglobulin gene polymorphisms of GM1, GM3, GM17, GM21 and KM3 in Mestizos [18]. Some risk genes were DM specific, such as BLK, IL-10 and TYK2 in Europeans [16,27,28], TNF-α and GSDMB in Caucasians [9,26], and PLCL1 in Caucasians and Han Chinese [16,29]. Some were PM-specific genes, such as IFN-γ in UK Caucasians [30], IFIH1 in Japanese [31], and SL26A1/IDUA in Caucasians [9]. Some genes were associated with serotype and clinical subgroups, such as TNF-α and anti-PM-Scl autoantibodies [26], IKBL (NFKBIL1), anti-Jo-1 and anti-PM-Scl autoantibodies [21], CCL21 and ILD [32].

Many genes also displayed protective roles in IIM, such as HLA-B*0702, HLA-Cw*0702, HLA-DRB1*0401, HLA-DRB1*0701 and HLA-DQA1*03 in North American Caucasians [14], GM21 in Koreans [33], CCL21 in DM in Caucasians [16], HLA-DRB1*0101, HLA-DRB1*1001 and HLA-DRB1*14 in PM in African-Americans [10], HLA-DRB1*0701, HLA-DRB1*01, HLA-DQA1*0101 and HLA-DQA1*0201 in anti-Jo-1 autoantibody-positive patients who were European Americans [17], and GM3 in anti-PL-7 and anti-RNP autoantibody-positive cases of European Americans [34].

Meaningful information has been acquired from GWASs, and these data may be useful for future research that leads to patient benefits in the clinic. Further research will require more homogenous cohorts and larger cohorts, as well as collaborations and trans ethnic meta-analyses between different research centres [35].

3. Epigenetics of DM/PM

As indicated by genome-wide complex trait analysis, SNPs explain 8.3% of the phenotypic variance in PM and 5.5% in DM, suggesting gene-specific epigenetic alterations in myositis [35]. Epigenetic mechanisms include DNA methylation, histone modification, and the activities of miRNA and long non-coding RNA. Here, we provide a comprehensive overview of microarray-based epigenetic modification in DM/PM patients.

DNA methylation is influenced both by the inherited DNA sequence and by environmental exposure [36], and it is considered to be a critical epigenetic factor that regulates gene expression by altering transcriptional chromatin accessibility of regulatory sequences within genes. The genome-global DNA methylation profiling of muscles revealed 27 genes, including homeobox and WT1 genes, with significant methylation differences between juvenile DM and normal controls, highlighting the important roles of DNA methylation in IIM [37]. In our study examining genome-level DNA methylation, we identified differentially methylated CpG sites in DM and PM, among which there are common and specifically methylated CpG site alterations between the two groups. In combination with gene expression profiles, DNA methylation modification may have a regulatory effect on approximately 20% of the differentially expressed genes, which enabled us to investigate the complexity of gene dysregulation in context (data not published).

MicroRNAs (miRNAs) play important roles in a wide range of developmental and physiological cellular processes in IIM. The expression levels and functions of miRNAs were widely studied in the serum, peripheral blood mononuclear cell (PBMC), skeletal muscles and skin of DM/PM patients. miRNAs are stable in serum and have potential use as biomarkers. Many dysregulated miRNAs were identified in the serum, including miR-223 and miR-7. These two miRNAs were decreased in DM and PM samples, in contrast to miR-31, which was increased [38]. miR-206 was decreased and miR-21 was increased in DM, and higher miR-21 levels were correlated with higher serum IgG levels, which may indicate an increased inflammatory burden [39]. miR-3907 levels were positively associated with disease activity in IIM [40]. In the PBMCs, miR-206 was decreased and augmented frequency of Th17 cells via Kruppel-like factor 4 [39]. In skeletal muscles, different miRNA expression patterns were found in DM and PM [41,42], miR-1, miR-133a and miR-133b were downregulated, and miR-146a, miR-221 and miR-222 were overexpressed; all of these miRNAs are involved in myocyte differentiation and muscle weakness. They also interact with cytokines. miR-146a also negatively regulates the type 1 IFN pathway [43]. miR-155 and miR-146b were increased and participated in the innate immune response. In DM lesion skin tissues, miR-7 was decreased and involved in inflammatory responses [44]. Decreased expression of miR-223 promoted keratinocyte growth via protein kinase Cε [45].

LncRNAs are RNA transcripts longer than 200 nucleotides with no coding potential. Aberrantly expressed lncRNAs in DM muscle tissues have been identified. Differential expression patterns of lncRNAs and distinct pathways were also found in DM with or without ILD or Jo-1 autoantibodies. linc-DGCR6-1 was highly expressed in the perifascicular areas of the muscle fibres and may regulate the type 1 interferon-inducible gene USP18. lncRNA studies open intriguing areas of inquiry for more in-depth exploration and to expand our understanding of the molecular mechanism of DM muscle damage [46].

Please cite this article as: Gao S, et al, Using multi-omics methods to understand dermatomyositis/polymyositis, Autoimmun Rev (2017), http://dx.doi.org/10.1016/j.autrev.2017.07.021

Given the complex aetiology of DM/PM, epigenetic studies are likely to provide new insights into the disease. Accumulated evidence has suggested that the interaction of genetic and environmental factors is required to initiate the onset of IIM symptoms [7]. A deeper understanding of the epigenetics of myositis will greatly help to clarify the pathogenesis of myositis and will help to identify new diagnostic biomarkers and therapeutic targets.

4. Transcriptomic profiles in DM/PM

Microarray-based transcription profiling is an attractive approach for understanding disease pathogenesis and for developing clinically relevant biomarkers. Global transcription profiling was performed on peripheral blood cells (PBCs), PBMCs, muscle biopsies and skin lesion tissues of DM/PM patients. Distinct gene profiles and type 1 interferon signatures were found in all of these studies. In PBCs, type 1 IFN-inducible gene transcript levels were highly correlated with disease activity; they decreased significantly after immunomodulatory therapies and showed concordant changes with major changes in disease activity [47–50]. In muscle biopsies, the genes that were differentially expressed were related to MHC I and II genes, cytokines, adhesion molecules, the actin cytoskeleton and immunoglobulins. Among the most altered genes, RSAD2 is an antiviral molecule that interacts with mitochondrial antiviral signalling protein [51] and is involved in endoplasmic reticulum stress in IIM [52]. S100A10 is a calcium-binding protein gene that participates in the dynamic function of muscle cells [53]. ISG15 is a ubiquitin-like modifier that conjugates with other proteins and was specifically found in DM muscle with perifascicular atrophy [54,55]. The chemokines CXCL9 and CXCL11 and several immunoglobulin-related genes were upregulated in the improved muscles of DM after immunoglobulin treatment, which helped us to identify patients who will respond to intravenous immunoglobulin infusion therapy [56]. In the DM skin lesion tissues, we also identified genes involved in inflammation, complement activation, and epidermal/kerationcyte activation and differentiation [57].

5. Proteomics in DM/PM

Proteomic techniques are important for identifying specific proteins, modification of proteins, and protein expression levels in different diseases states, and they can be used to characterize subtly different disease phenotypes. Proteomic studies have been conducted in DM/PM muscle biopsies, bronchoalveolar lavage fluid and circulating immune complexes [54,58,59]. In the DM perifascicular atrophic myofibres, the ISG15-conjugation pathway proteins ISG15, HERC5, USP18, MxA, IFIT1 and IFIT3 were markedly upregulated, and titin was downregulated [54]. The pulmonary parenchyma is frequently involved in IIM, and the protein profiles of bronchoalveolar lavage fluid (BALF) samples may constitute a window into the molecules that mediate interactions between IIM and lung disease. From the highly expressed proteins (323 ± 51 proteins) in BALF, 9 spots were PM/DM specific, including the cytoskeleton/tissue architecture proteins; however, no correlations with disease activity were observed [58]. Immune complexes (ICs) trigger humoral immune responses, and the identification of constituent antigens within ICs would have more clinical significance than the identification of free antigens. In the DM serum ICs, 6 IgG and 19 IgA disease-specific antigens were identified and may become promising diagnostic or pathogenic biomarkers [59]. Proteomic analysis was also performed on the mitochondrial proteins from muscle samples from PM patients; the results showed that electron transport chain complex subunits, assembly factors and Krebs cycle enzymes were downregulated, and increased tryptophan (Trp) oxidation was also observed. Altered mitochondrial proteome and protein oxidation established the important roles of mitochondrial damage in human muscle diseases [60].

Technical developments in proteomics have greatly advanced protein identification, biochemical analysis and expression profiling. In DM/PM, proteomics is essential for the design of improved diagnostics, the monitoring of pathophysiological progression, the evaluation of pharmacological action and the identification of novel therapeutic targets.

6. Autoantibodies in DM/PM

Autoantibodies are important biomarkers in IIM. In recent years, enthusiasm and efforts have been devoted to finding new autoantibodies in IIM patients. Several autoantibodies are specific to IIM, known as myositis-specific autoantibodies (MSAs), and others are associated with the myositis overlap syndrome, known as myositis-associated autoantibodies (MAAs). MSAs are closely correlated with the clinical phenotype, complications, reactivity to therapy, and prognosis of the disease [61,62].

Anti-ARSs are a group of autoantibodies that recognize the cytoplasmic amino acid-charging enzymes, including histidyl (Jo-1), threonyl (PL-7), alanyl (PL-12), glycyl (EJ), isoleucyl (OJ), asparaginyl (KS), phenylalanyl (ZO), and tyrosyl (YRS/HA) tRNA synthetases [63]. The anti-Jo-1 autoantibody was the first MSA to be discovered and by far the most common anti-ARS [64], and is usually found in 15–25% of PM/DM patients, while all other anti-ARSs are usually found only in 0.5–6% of patients [65,66]. Anti-ARS is associated with antisynthetase syndrome, which consists of myopathy, fever, ILD, Raynaud's phenomenon, arthritis and “mechanic's hands”. Patients negative for anti-Jo-1 autoantibodies are more likely to have earlier and more severe ILD without typical myositis and a poor prognosis compared with anti-Jo-1 (+) patients [67].

The anti-signal recognition particle (anti-SRP) autoantibody is directed against a polypeptide complex that is involved in protein transport to the endoplasmic reticulum. Anti-SRP is specific to PM and is found in 4% of PM/DM patients, and in 0–54% in patients with necrotizing myopathy. Although the anti-SRP antibody is not considered directly pathogenic, it is associated with treatment-resistant severe myopathy [68–70].

Anti-Mi-2 is an antibody for a nuclear DNA helicase involved in transcriptional activation. The prevalence of the anti-Mi-2 autoantibody is 2–50% in DM and 0–10% in PM [71]. Anti-Mi-2 is associated with typical skin lesions of DM, but the risk of developing clinically significant ILD is low, and cancer is uncommon [19,72,73].

Anti-melanoma differentiation-associated protein-5/CADM140 (anti-MDA-5/CADM140) is directed against a cytoplasmic RNA-specific helicase [8]. This autoantibody is specific to DM and showed a prevalence ranging from 3 to 58%; however, the prevalence increased up to 100% when only CADM (typical DM skin rash but with little or no muscle involvement) patients were considered [74]. Anti-MDA5/CADM140 was associated with CADM or rapidly progressive ILD, a high prevalence of liver dysfunction, and increased serum levels of ferritin [75].

Anti-TIF1γ/α and β is an autoantibody referred to as anti-p155/140. It immunoprecipitates a set of 155 and 140 kD proteins [76,77]. TIF1α, TIF1β and TIF1γ belong to the TIF family of transcription cofactors and are part of a tripartite motif superfamily [78]. The prevalence of anti-TIF1γ/α in cancer-associated DM is 22–100%, and anti-TIF1γ/α has a striking association with cancer-associated DM [79]. However, it should be noted that the association of anti-TIF1γ/α with cancer does not seem to apply to children or young adults affected by DM [80].

Anti-MJ antibodies recognize an ~140-kD nuclear protein called nuclear matrix protein 2 (NXP-2), which plays an important role in diverse nuclear functions such as RNA metabolism and the maintenance of nuclear architecture [81]. The prevalence of anti-MJ/NXP-2 autoantibodies can be very different in studies performed worldwide, which could be due to different ethnic backgrounds, the influence of environmental factors on autoantibody production, or simply for technical reasons [79]. The clinical association of anti-MJ/NXP-2 was also varied.

The target antigens of SAE are heterodimer proteins including small ubiquitin-like modifier-activating enzyme A subunit (SAE1) and the

Please cite this article as: Gao S, et al, Using multi-omics methods to understand dermatomyositis/polymyositis, Autoimmun Rev (2017), http://dx.doi.org/10.1016/j.autrev.2017.07.021

SUMO-1 activating enzyme B subunit (SAE2), which are enzymes involved in the post-translational modification of specific proteins known as SUMOylation [82]. Anti-SAE was found in 4% of DM/PM patients and in 7–10% of DM patients. Anti-SAE was associated with cutaneous lesions, including heliotrope, gottron rash, dysphagia and skin disease prior to the onset of myositis [83,84].

Recently, significant progress in MSA has been made, and several new MSAs with strong clinical significance have been identified. New immunoassays for these new MSAs, as well as classic MSAs will become widely available for clinical practice in the near future.

7. Discussion

In contrast to conventional hypothesis-driven bioresearch that routinely focuses on a select number of biomolecular factors, multi-omics research attempts to evaluate global changes in disease progression. The integrated analysis of genomic, epigenetic, transcriptomic and proteomic profiling of entire tissue systems will provide useful insights into DM/PM pathogenesis and will provide direction for therapeutic targets and biomarkers development. To date, most studies have used only a single omics approach and thus have failed to fully discover the complex mechanisms of DM/PM. No study or review has completely analysed the pathogenesis of DM/PM using a multi-omics approach. This review is the first to combine all omics studies from genomics to autoantibodies and may serve as a reference for future studies to identify the key underlying pathways in DM/PM. Integrated multi-omics data may facilitate the elucidation of functional principles at different levels, such as genes that are regulated at the chromatin, transcriptional, post-transcriptional, translational and post-translational levels in the coordinated progression of the diseases process. Due to the diversity of DM/PM, it is important to expand the use of multi-omics and systems biology methods to facilitate a more precise understanding of the mechanistic principles underlying DM/PM and to improve treatment in the future.

References

[1] Tieu J, Lundberg IE, Limaye V. Idiopathic inflammatory myositis. Best Pract Res Clin Rheumatol 2016;30:149–68.
[2] Dimachkie MM, Barohn RJ, Amato AA. Idiopathic inflammatory myopathies. Neurol Clin 2014;32:595–628 [vii].
[3] Dalakas MC. Inflammatory muscle diseases. N Engl J Med 2015;372:1734–47.
[4] Ernste FC, Reed AM. Idiopathic inflammatory myopathies: current trends in pathogenesis, clinical features, and up-to-date treatment recommendations. Mayo Clin Proc 2013;88:83–105.
[5] Ceribelli A, De Santis M, Isailovic N, Gershwin ME, Selmi C. The immune response and the pathogenesis of idiopathic inflammatory myositis: a critical review. Clin Rev Allergy Immunol 2017;52:58–70.
[6] Mammen AL. Autoimmune myopathies: autoantibodies, phenotypes and pathogenesis. Nat Rev Neurol 2011;7:343–54.
[7] Gan L, O'Hanlon TP, Gordon AS, Rider LG, Miller FW, Burbelo PD. Twins discordant for myositis and systemic lupus erythematosus show markedly enriched autoantibodies in the affected twin supporting environmental influences in pathogenesis. BMC Musculoskelet Disord 2014;15:67.
[8] Miller FW, Chen W, O'Hanlon TP, Cooper RG, Vencovsky J, Rider LG, et al. Genome-wide association study identifies HLA 8.1 ancestral haplotype alleles as major genetic risk factors for myositis phenotypes. Genes Immun 2015;16:470–80.
[9] Rothwell S, Cooper RG, Lundberg IE, Miller FW, Gregersen PK, Bowes J, et al. Dense genotyping of immune-related loci in idiopathic inflammatory myopathies confirms HLA alleles as the strongest genetic risk factor and suggests different genetic background for major clinical subgroups. Ann Rheum Dis 2016;75:1558–66.
[10] O'Hanlon TP, Rider LG, Mamyrova G, Targoff IN, Arnett FC, Reveille JD, et al. HLA polymorphisms in African Americans with idiopathic inflammatory myopathy: allelic profiles distinguish patients with different clinical phenotypes and myositis autoantibodies. Arthritis Rheum 2006;54:3670–81.
[11] Chinoy H, Salway F, Fertig N, Shephard N, Tait BD, Thomson W, et al. In adult onset myositis, the presence of interstitial lung disease and myositis specific/associated antibodies are governed by HLA class II haplotype, rather than by myositis subtype. Arthritis Res Ther 2006;8:R13.
[12] Chinoy H, Payne D, Poulton KV, Fertig N, Betteridge Z, Gunawardena H, et al. HLA-DPB1 associations differ between DRB1*03 positive anti-Jo-1 and anti-PM-Scl antibody positive idiopathic inflammatory myopathy. Rheumatology 2009;48:1213–7.
[13] Furuya T, Hakoda M, Tsuchiya N, Kotake S, Ichikawa N, Nanke Y, et al. Immunogenetic features in 120 Japanese patients with idiopathic inflammatory myopathy. J Rheumatol 2004;31:1768–74.
[14] O'Hanlon TP, Carrick DM, Arnett FC, Reveille JD, Carrington M, Gao X, et al. Immunogenetic risk and protective factors for the idiopathic inflammatory myopathies: distinct HLA-A, -B, -Cw, DRB1 and -DQA1 allelic profiles and motifs define clinicopathologic groups in caucasians. Medicine 2005;84:338–49.
[15] Gao X, Han L, Yuan L, Yang Y, Gou G, Sun H, et al. HLA class II alleles may influence susceptibility to adult dermatomyositis and polymyositis in a Han Chinese population. BMC Dermatol 2014;14:9.
[16] Miller FW, Cooper RG, Vencovsky J, Rider LG, Danko K, Wedderburn LR, et al. Genome-wide association study of dermatomyositis reveals genetic overlap with other autoimmune disorders. Arthritis Rheum 2013;65:3239–47.
[17] O'Hanlon TP, Carrick DM, Targoff IN, Arnett FC, Reveille JD, Carrington M, et al. Immunogenetic risk and protective factors for the idiopathic inflammatory myopathies: distinct HLA-A, -B, -Cw, -DRB1, and -DQA1 allelic profiles distinguish European American patients with different myositis autoantibodies. Medicine 2006;85:111–27.
[18] Shamim EA, Rider LG, Pandey JP, O'Hanlon TP, Jara LJ, Samayoa EA, et al. Differences in idiopathic inflammatory myopathy phenotypes and genotypes between Mesoamerican Mestizos and North American Caucasians: ethnogeographic influences in the genetics and clinical expression of myositis. Arthritis Rheum 2002;46:1885–93.
[19] Hausmanowa-Petrusewicz I, Kowalska-Oledzka E, Miller FW, Jarzabek-Chorzelska M, Targoff IN, Blaszczyk-Kostanecka M, et al. Clinical, serologic, and immunogenetic features in Polish patients with idiopathic inflammatory myopathies. Arthritis Rheum 1997;40:1257–66.
[20] Chinoy H, Salway F, John S, Fertig N, Tait BD, Oddis CV, et al. Tumour necrosis factor-alpha single nucleotide polymorphisms are not independent of HLA class I in UK Caucasians with adult onset idiopathic inflammatory myopathies. Rheumatology 2007;46:1411–6.
[21] Chinoy H, Li CK, Platt H, Fertig N, Varsani H, Gunawardena H, et al. Genetic association study of NF-kappaB genes in UK Caucasian adult and juvenile onset idiopathic inflammatory myopathy. Rheumatology 2012;51:794–9.
[22] Chinoy H, Platt H, Lamb JA, Betteridge Z, Gunawardena H, Fertig N, et al. The protein tyrosine phosphatase N22 gene is associated with juvenile and adult idiopathic inflammatory myopathy independent of the HLA 8.1 haplotype in British Caucasian patients. Arthritis Rheum 2008;58:3247–54.
[23] Sandoval-Garcia F, Petri MH, Saavedra MA, Cruz-Reyes CV, Jara-Quezada LJ, Davalos-Rodriguez IP, et al. The ACTN3 R577X polymorphism is associated with inflammatory myopathies in a Mexican population. Scand J Rheumatol 2012;41:396–400.
[24] Sugiura T, Kawaguchi Y, Goto K, Hayashi Y, Gono T, Furuya T, et al. Association between a C8orf13-BLK polymorphism and polymyositis/dermatomyositis in the Japanese population: an additive effect with STAT4 on disease susceptibility. PLoS One 2014;9:e90019.
[25] Sugiura T, Kawaguchi Y, Goto K, Hayashi Y, Tsuburaya R, Furuya T, et al. Positive association between STAT4 polymorphisms and polymyositis/dermatomyositis in a Japanese population. Ann Rheum Dis 2012;71:1646–50.
[26] Chen S, Wang Q, Wu Z, Li Y, Li P, Sun F, et al. Genetic association study of TNFAIP3, IFIH1, IRF5 polymorphisms with polymyositis/dermatomyositis in Chinese Han population. PLoS One 2014;9:e110044.
[27] Hristova M, Dourmishev L, Kamenarska Z, Kaneva R, Vinkov A, Mitev V. Association of tumor necrosis factor alpha (TNF-alpha) and interleukin 10 (IL-10) gene polymorphisms in dermatomyositis patients: a pilot study. Acta Dermatovenerol Croat 2012;20:148–56.
[28] Jani M, Massey J, Wedderburn LR, Vencovsky J, Danko K, Lundberg IE, et al. Genotyping of immune-related genetic variants identifies TYK2 as a novel associated locus for idiopathic inflammatory myopathies. Ann Rheum Dis 2014;73:1750–2.
[29] Wang Q, Chen S, Li Y, Li P, Wu C, Wu Z, et al. Positive association of genetic variations in the phospholipase C-like 1 gene with dermatomyositis in Chinese Han. Immunol Res 2016;64:204–12.
[30] Chinoy H, Salway F, John S, Fertig N, Tait BD, Oddis CV, et al. Interferon-gamma and interleukin-4 gene polymorphisms in Caucasian idiopathic inflammatory myopathy patients in UK. Ann Rheum Dis 2007;66:970–3.
[31] Gono T, Kawaguchi Y, Sugiura T, Furuya T, Kawamoto M, Hanaoka M, et al. Interferon-induced helicase (IFIH1) polymorphism with systemic lupus erythematosus and dermatomyositis/polymyositis. Mod Rheumatol 2010;20:466–70.
[32] Chen S, Wang Q, Wu CY, Wu QJ, Li Y, Wu ZY, et al. A single-nucleotide polymorphism of CCL21 rs951005 T>C is associated with susceptibility of polymyositis and such patients with interstitial lung disease in a Chinese Han population. Clin Exp Rheumatol 2015;33:639–46.
[33] Rider LG, Shamim E, Okada S, Pandey JP, Targoff IN, O'Hanlon TP, et al. Genetic risk and protective factors for idiopathic inflammatory myopathy in Koreans and American whites: a tale of two loci. Arthritis Rheum 1999;42:1285–90.
[34] O'Hanlon TP, Rider LG, Schiffenbauer A, Targoff IN, Malley K, Pandey JP, et al. Immunoglobulin gene polymorphisms are susceptibility factors in clinical and autoantibody subgroups of the idiopathic inflammatory myopathies. Arthritis Rheum 2008;58:3239–46.
[35] Rothwell S, Lamb JA, Chinoy H. New developments in genetics of myositis. Curr Opin Rheumatol 2016;28:651–6.
[36] Carrio E, Suelves M. DNA methylation dynamics in muscle development and disease. Front Aging Neurosci 2015;7:19.
[37] Wang M, Xie H, Shrestha S, Sredni S, Morgan GA, Pachman LM. Methylation alterations of WT1 and homeobox genes in inflamed muscle biopsy samples from patients with untreated juvenile dermatomyositis suggest self-renewal capacity. Arthritis Rheum 2012;64:3478–85.
[38] Zhou B, Zuo XX, Li YS, Gao SM, Dai XD, Zhu HL, et al. Integration of microRNA and mRNA expression profiles in the skin of systemic sclerosis patients. Sci Rep 2017;7:42899.
[39] Tang X, Tian X, Zhang Y, Wu W, Tian J, Rui K, et al. Correlation between the frequency of Th17 cell and the expression of microRNA-206 in patients with dermatomyositis. Clin Dev Immunol 2013;2013:345347.

Please cite this article as: Gao S, et al, Using multi-omics methods to understand dermatomyositis/polymyositis, Autoimmun Rev (2017), http://dx.doi.org/10.1016/j.autrev.2017.07.021

ARTICLE IN PRESS

[40] Misunova M, Salinas-Riester G, Luthin S, Pommerenke C, Husakova M, Zavada J, et al. Microarray analysis of circulating micro RNAs in the serum of patients with polymyositis and dermatomyositis reveals a distinct disease expression profile and is associated with disease activity. Clin Exp Rheumatol 2016;34:17–24.
[41] Eisenberg I, Eran A, Nishino I, Moggio M, Lamperti C, Amato AA, et al. Distinctive patterns of microRNA expression in primary muscular disorders. Proc Natl Acad Sci U S A 2007;104:17016–21.
[42] Georgantas RW, Streicher K, Greenberg SA, Greenlees LM, Zhu W, Brohawn PZ, et al. Inhibition of myogenic microRNAs 1, 133, and 206 by inflammatory cytokines links inflammation and muscle degeneration in adult inflammatory myopathies. Arthritis Rheumatol 2014;66:1022–33.
[43] Zhu W, Streicher K, Shen N, Higgs BW, Morehouse C, Greenlees L, et al. Genomic signatures characterize leukocyte infiltration in myositis muscles. BMC Med Genet 2012;5:53.
[44] Oshikawa Y, Jinnin M, Makino T, Kajihara I, Makino K, Honda N, et al. Decreased miR-7 expression in the skin and sera of patients with dermatomyositis. Acta Derm Venereol 2013;93:273–6.
[45] Inoue K, Jinnin M, Yamane K, Makino T, Kajihara I, Makino K, et al. Down-regulation of miR-223 contributes to the formation of Gottron's papules in dermatomyositis via the induction of PKCvarepsilon. Eur J Dermatol 2013;23:160–7.
[46] Peng QL, Zhang YM, Yang HB, Shu XM, Lu X, Wang GC. Transcriptomic profiling of long non-coding RNAs in dermatomyositis by microarray analysis. Sci Rep 2016;6:32818.
[47] Baechler EC, Bauer JW, Slattery CA, Ortmann WA, Espe KJ, Novitzke J, et al. An interferon signature in the peripheral blood of dermatomyositis patients is associated with disease activity. Mol Med 2007;13:59–68.
[48] Liao AP, Salajegheh M, Nazareno R, Kagan JC, Jubin RG, Greenberg SA. Interferon beta is associated with type 1 interferon-inducible gene expression in dermatomyositis. Ann Rheum Dis 2011;70:831–6.
[49] Greenberg SA, Higgs BW, Morehouse C, Walsh RJ, Kong SW, Brohawn P, et al. Relationship between disease activity and type 1 interferon- and other cytokine-inducible gene expression in blood in dermatomyositis and polymyositis. Genes Immun 2012;13:207–13.
[50] Walsh RJ, Kong SW, Yao Y, Jallal B, Kiener PA, Pinkus JL, et al. Type I interferon-inducible gene expression in blood is present and reflects disease activity in dermatomyositis and polymyositis. Arthritis Rheum 2007;56:3784–92.
[51] Seo JY, Yaneva R, Cresswell P. Viperin: a multifunctional, interferon-inducible protein that regulates virus replication. Cell Host Microbe 2011;10:534–9.
[52] Suarez-Calvet X, Gallardo E, Nogales-Gadea G, Querol L, Navas M, Diaz-Manera J, et al. Altered RIG-I/DDX58-mediated innate immunity in dermatomyositis. J Pathol 2014;233:258–68.
[53] Zhou X, Dimachkie MM, Xiong M, Tan FK, Arnett FC. cDNA microarrays reveal distinct gene expression clusters in idiopathic inflammatory myopathies. Med Sci Monit 2004;10:BR191–7.
[54] Salajegheh M, Kong SW, Pinkus JL, Walsh RJ, Liao A, Nazareno R, et al. Interferon-stimulated gene 15 (ISG15) conjugates proteins in dermatomyositis muscle with perifascicular atrophy. Ann Neurol 2010;67:53–63.
[55] Gerami P, Schope JM, McDonald L, Walling HW, Sontheimer RD. A systematic review of adult-onset clinically amyopathic dermatomyositis (dermatomyositis sine myositis): a missing link within the spectrum of the idiopathic inflammatory myopathies. J Am Acad Dermatol 2006;54:597–613.
[56] Raju R, Dalakas MC. Gene expression profile in the muscles of patients with inflammatory myopathies: effect of therapy with IVIg and biological validation of clinically relevant genes. Brain J Neurol 2005;128:1887–96.
[57] Wong D, Kea B, Pesich R, Higgs BW, Zhu W, Brown P, et al. Interferon and biologic signatures in dermatomyositis skin: specificity and heterogeneity across diseases. PLoS One 2012;7:e29161.
[58] Passadore I, Iadarola P, Di Poto C, Giuliano S, Montecucco C, Cavagna L, et al. 2-DE and LC-MS/MS for a comparative proteomic analysis of BALf from subjects with different subsets of inflammatory myopathies. J Proteome Res 2009;8:2331–40.
[59] Ohyama K, Baba M, Tamai M, Aibara N, Ichinose K, Kishikawa N, et al. Proteomic profiling of antigens in circulating immune complexes associated with each of seven autoimmune diseases. Clin Biochem 2015;48:181–5.
[60] Sunitha B, Gayathri N, Kumar M, Keshava Prasad TS, Nalini A, Padmanabhan B, et al. Muscle biopsies from human muscle diseases with myopathic pathology reveal common alterations in mitochondrial function. J Neurochem 2016;138:174–91.
[61] Ghirardello A, Bassi N, Palma L, Borella E, Domeneghetti M, Punzi L, et al. Autoantibodies in polymyositis and dermatomyositis. Curr Rheumatol Rep 2013;15:335.
[62] Lu X, Peng Q, Wang G. Discovery of new biomarkers of idiopathic inflammatory myopathy. Clin Chim Acta 2015;444:117–25.
[63] Jura M, Rychlewski L, Barciszewski J. Comprehensive insight into human aminoacyl-tRNA synthetases as autoantigens in idiopathic inflammatory myopathies. Crit Rev Immunol 2007;27:559–72.
[64] Nishikai M, Reichlin M. Heterogeneity of precipitating antibodies in polymyositis and dermatomyositis. Characterization of the Jo-1 antibody system. Arthritis Rheum 1980;23:881–8.
[65] Targoff IN. Update on myositis-specific and myositis-associated autoantibodies. Curr Opin Rheumatol 2000;12:475–81.
[66] Targoff IN, Arnett FC, Reichlin M. Antibody to threonyl-transfer RNA synthetase in myositis sera. Arthritis Rheum 1988;31:515–24.
[67] Aggarwal R, Cassidy E, Fertig N, Koontz DC, Lucas M, Ascherman DP, et al. Patients with non-Jo-1 anti-tRNA-synthetase autoantibodies have worse survival than Jo-1 positive patients. Ann Rheum Dis 2014;73:227–32.
[68] Kao AH, Lacomis D, Lucas M, Fertig N, Oddis CV. Anti-signal recognition particle autoantibody in patients with and patients without idiopathic inflammatory myopathy. Arthritis Rheum 2004;50:209–15.
[69] Hengstman GJ, ter Laak HJ, Vree Egberts WT, Lundberg IE, Moutsopoulos HM, Vencovsky J, et al. Anti-signal recognition particle autoantibodies: marker of a necrotising myopathy. Ann Rheum Dis 2006;65:1635–8.
[70] Miller T, Al-Lozi MT, Lopate G, Pestronk A. Myopathy with antibodies to the signal recognition particle: clinical and pathological features. J Neurol Neurosurg Psychiatry 2002;73:420–8.
[71] Targoff IN, Reichlin M. The association between Mi-2 antibodies and dermatomyositis. Arthritis Rheum 1985;28:796–803.
[72] Love LA, Leff RL, Fraser DD, Targoff IN, Dalakas M, Plotz PH, et al. A new approach to the classification of idiopathic inflammatory myopathy: myositis-specific autoantibodies define useful homogeneous patient groups. Medicine 1991;70:360–74.
[73] Brouwer R, Hengstman GJ, Vree Egberts W, Ehrfeld H, Bozic B, Ghirardello A, et al. Autoantibody profiles in the sera of European patients with myositis. Ann Rheum Dis 2001;60:116–23.
[74] Petri MH, Satoh M, Martin-Marquez BT, Vargas-Ramirez R, Jara LJ, Saavedra MA, et al. Implications in the difference of anti-Mi-2 and -p155/140 autoantibody prevalence in two dermatomyositis cohorts from Mexico City and Guadalajara. Arthritis Res Ther 2013;15:R48.
[75] Gono T, Kawaguchi Y, Satoh T, Kuwana M, Katsumata Y, Takagi K, et al. Clinical manifestation and prognostic factor in anti-melanoma differentiation-associated gene 5 antibody-associated interstitial lung disease as a complication of dermatomyositis. Rheumatology 2010;49:1713–9.
[76] Targoff IN, Mamyrova G, Trieu EP, Perurena O, Koneru B, O'Hanlon TP, et al. A novel autoantibody to a 155-kd protein is associated with dermatomyositis. Arthritis Rheum 2006;54:3682–9.
[77] Satoh M, Chan JY, Ross SJ, Li Y, Yamasaki Y, Yamada H, et al. Autoantibodies to transcription intermediary factor TIF1beta associated with dermatomyositis. Arthritis Res Ther 2012;14:R79.
[78] Herquel B, Ouararhni K, Khetchoumian K, Ignat M, Teletin M, Mark M, et al. Transcription cofactors TRIM24, TRIM28, and TRIM33 associate to form regulatory complexes that suppress murine hepatocellular carcinoma. Proc Natl Acad Sci U S A 2011;108:8212–7.
[79] Satoh M, Tanaka S, Ceribelli A, Calise SJ, Chan EK. A comprehensive overview on myositis-specific antibodies: new and old biomarkers in idiopathic inflammatory myopathy. Clin Rev Allergy Immunol 2017;52:1–19.
[80] Trallero-Araguas E, Labrador-Horrillo M, Selva-O'Callaghan A, Martinez MA, Martinez-Gomez X, Palou E, et al. Cancer-associated myositis and anti-p155 autoantibody in a series of 85 patients with idiopathic inflammatory myopathy. Medicine 2010;89:47–52.
[81] Mimura Y, Takahashi K, Kawata K, Akazawa T, Inoue N. Two-step colocalization of MORC3 with PML nuclear bodies. J Cell Sci 2010;123:2014–24.
[82] Betteridge Z, Gunawardena H, North J, Slinn J, McHugh N. Identification of a novel autoantibody directed against small ubiquitin-like modifier activating enzyme in dermatomyositis. Arthritis Rheum 2007;56:3132–7.
[83] Tarricone E, Ghirardello A, Rampudda M, Bassi N, Punzi L, Doria A. Anti-SAE antibodies in autoimmune myositis: identification by unlabelled protein immunoprecipitation in an Italian patient cohort. J Immunol Methods 2012;384:128–34.
[84] Fujimoto M, Matsushita T, Hamaguchi Y, Kaji K, Asano Y, Ogawa F, et al. Autoantibodies to small ubiquitin-like modifier activating enzymes in Japanese patients with dermatomyositis: comparison with a UK Caucasian cohort. Ann Rheum Dis 2013;72:151–3.

Please cite this article as: Gao S, et al, Using multi-omics methods to understand dermatomyositis/polymyositis, Autoimmun Rev (2017), http://dx.doi.org/10.1016/j.autrev.2017.07.021